

Streamlined Numerical Working Note and Experiment Plan

Asadullah Bhuiyan¹

¹*Department of Physics, Cornell University, Ithaca, New York 14853, USA*

(Dated: September 10, 2026)

This living note is an analysis-first numerical plan for the adaptive class-A fermion-circuit paper. It replaces the assumption that every claim must be regenerated on one new master ensemble. The repository already contains strong exact and stochastic evidence for bulk topology, a critical wall state, a wall-localized slow tangent sector, wall-localized purification, and modular packet handedness. The immediate task is to analyze those data consistently and run only the experiments needed to close a retained paper claim. New Colab work uses deterministic tasks, visible `tqdm` progress, completion-based resume, and one rolling scientific checkpoint only for unusually long tasks. The current modular-handedness extension is a local CPU campaign rather than a Colab campaign.

I. PURPOSE AND WORKING RULES

This is the single working experiment note to update as the paper scope changes. It separates three questions:

1. What do the saved arrays already establish?
2. What analysis is missing although the simulation is complete?
3. What new simulation is required by the final manuscript claim?

The default rule is *analyze before rerunning*. A legacy result is not accepted merely because a figure exists, but an auditable raw archive is not discarded because it pre-dates the latest runner. Protocol, geometry, initialization, schedule, estimator, fit window, and source version must be recorded. A small same-seed equivalence test is preferred over a full rerun when the only concern is engine provenance.

Exact Hamiltonian, mean-channel, stochastic Born-trajectory, postselected, and modular-flow results remain separately labelled. They may serve as calibration or controls but are not pooled as measurements of the same object.

II. EXISTING EVIDENCE AND NEXT ACTIONS

The starting artifacts are the exact B0 audit, frozen P1 production, the joint-observable $S = 10$ archive, recovered endpoint archives, legacy Lyapunov and purification campaigns, and the mean-channel production matrix. Their locations are catalogued in `PROJECT_ADMIN/DATA_CATALOG.md` and the legacy evidence atlas. This note should cite exact paths as each analysis is completed rather than duplicating a second data catalog.

III. IMMEDIATE ANALYSIS QUEUE: NO NEW GPU TIME

A. A0: uniform bulk-validation closure

On September 9, 2026, the uniform perfect-correction validation campaign was declared scientifically complete on its balanced $L = 12, 16, 20, 24, 28, 32$ grid. For each size and $n_{\text{shell}} = 1, 2$, dense, the dataset contains $S = 100$ independent pure half-filled trajectories, no domain wall, 40 random-serial perfect-correction cycles, complex128 arithmetic, and the trajectory-level real-space Chern and charge estimators at every cycle including zero. Thus the accepted comparison contains 18 cases and 1800 trajectories. All 18 cycle-40 ensemble means lie between 0.99556 and 0.9999996.

The planned completion of $L = 36, 40$ is retired. Existing partial $L = 36$ outputs remain preserved as supporting data but are excluded from the balanced headline size series; no $L = 40$ result is claimed. The exact decision and data boundary are recorded in `experiment_review/uniform_bulk_validation_analysis/CAMPAIGN_CLOSURE.md`.

B. A1: joint wall-state reduction

Analyze the completed random-serial pure-state archive at $N_x = 20$ and $N_y = 20, 30, 40, 50$. Use per-trajectory estimators followed by equal-weight trajectory averaging. Report:

1. entropy and charge-fluctuation slopes on identical interval windows;
2. c and k with trajectory-bootstrap intervals;
3. explicit-interface versus support-terminated construction differences;
4. matched-trivial null results;
5. fit-window and late-time sensitivity;
6. Chern, stationarity, and numerical-quality diagnostics.

TABLE I. Initial evidence map. The last column gives the smallest responsible next action rather than a new master campaign.

Object	Existing evidence	Safe present claim	Next action
Exact benchmark	wall Twenty-two audited geometries; $c \simeq 1$, correlator exponent near 2.074, opposite physical and modular velocities, and integer twist flow	The target is a chiral $c = 1$ free-fermion wall, with accepted $N_x = 20$	Reuse as calibration; do not rerun
Bulk topology	stochastic Frozen production has 48 cases, 240 archives, and 1200 complex128 trajectories with topological and trivial controls. The newer every-cycle uniform campaign adds a balanced 1800-trajectory grid at $L = 12$ –32 for $n_{\text{shell}} = 1, 2$, dense	The tested adaptive circuit prepares the intended bulk phases	Bulk validation is complete; do not run redesigned P1 or the retired $L = 36, 40$ top-up
Wall-state CFT data	Existing $S = 100$ entropy/correlator data and a separate $S = 10$ archive with entropy, charge fluctuations, correlations, Chern data, tangent modes, and records on the same trajectories	The tested wall is consistent with $c \approx k \approx 1$ and algebraic correlations	Formally analyze the joint $S = 10$ archive before choosing a top-up
Tangent sector	CPU and GPU campaigns show a small wall gap, a much larger trivial gap, and approximately 96–97% wall-plus-neighbour weight	A finite-size slow tangent sector is strongly localized	Extend only if a thermodynamic zero-gap claim remains
Purification	$S = 100$ max-mixed trajectory; approximately 93–94% of the late residual entropy lies on wall columns	Purification is fast in the bulk and leaves a wall-localized residual	Reanalyze completed width data; do not rerun a broad matrix
Chiral packet motion	Legacy packets and recovered endpoint archives show opposite wall displacements; hard topological has 20 trajectories and hard trivial has 25	Modular packet transport is handed and absent in the matched-trivial control	Resolve absolute sign; run the launched fixed-geometry CPU ensemble
Exact flux and mean channel	Stable exact twist/pump calculations and a completed channel matrix	Exact and unconditional controls have oriented mean-wall response	Reuse as supplemental calibration, not sampled Born response

Only after A1 may a specific size or window be selected for a top-up.

1. Power-law scaling of the x -averaged correlator

Here is the critical prediction. A chiral complex fermion has $G(r_y) \sim d_{N_y}(r_y)^{-1}$, so the squared correlator measured in the circuit must fall with exponent two,

$$d_{N_y}(r_y) = \frac{N_y}{\pi} \sin\left(\frac{\pi r_y}{N_y}\right), \quad (1)$$

$$C_G(r_y) \sim d_{N_y}(r_y)^{-\beta}, \quad \beta_{\text{crit}} = 2.$$

That is the number we want to see emerge; it is not imposed by the fit.

For trajectory ξ , first form the fixed- x squared Green function

$$C_{G,\xi}(x, r_y) = \frac{1}{2N_y} \sum_{y,\mu,\nu} |C_{\xi;(x,y,\mu),(x,y+r_y,\nu)}|^2, \quad (2)$$

and then average across the transverse direction,

$$C_{G,\xi}^{\text{av}}(r_y) = \frac{1}{N_x} \sum_x C_{G,\xi}(x, r_y). \quad (3)$$

The x average mixes the two algebraic wall pieces with short-ranged bulk pieces. The bulk contribution falls away quickly, leaving the wall power law at the distances used here. We fit each trajectory on $2 \leq r_y \leq \lfloor N_y/4 \rfloor$, then summarize the resulting β_ξ across the $S = 100$ trajectories.

Figure 1(a) is the direct test: all six circumferences follow the same algebraic envelope, close to the d^{-2} prediction. Panel (b) asks the finite-size question in the same form used elsewhere in this note. The measured exponent drifts from roughly 2.29 at the smaller circumferences to 2.23 at the largest. Extrapolations linear in $1/N_y$ and $1/N_y^2$ land at 2.18 and 2.23, respectively. The important result is therefore simple: the stochastic wall has the predicted algebraic sector and is moving toward two, while the remaining drift is still large enough that we should call this consistency with $\beta = 2$, not a precision measurement of the thermodynamic exponent.

The x -resolved archives let us make the sharper test directly on the two programmed wall columns. Repeating exactly the same fit at $x_L = 5$ and $x_R = 15$ gives $\beta_L = 2.17$ –2.18 and $\beta_R = 2.27$ –2.28 across $N_y = 24, 28, 32$. Averaging the two wall curves within each trajectory gives $\beta_{\text{wall}} = 2.216$ –2.223. These values barely move with cir-

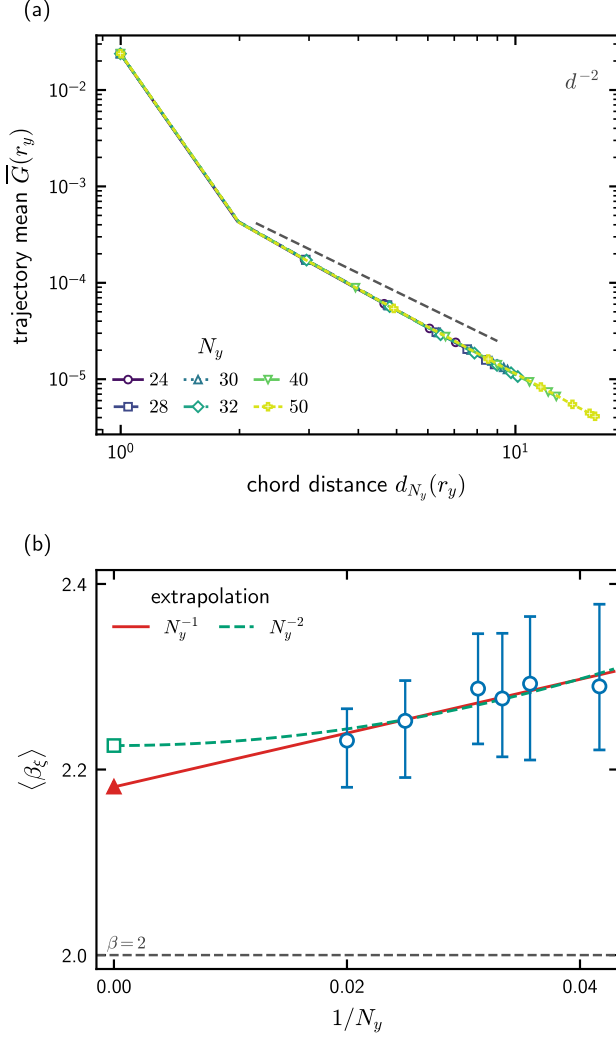


FIG. 1. Hard-wall correlator scaling at $N_x = 20$. (a) Trajectory- and x -averaged squared correlator after $2N_y$ perfect-correction cycles; the dashed segment shows the critical d^{-2} prediction. (b) Mean trajectory exponent versus inverse circumference. Each size contains $S = 100$ independent pure half-filled trajectories. Fits use $2 \leq r_y \leq \lfloor N_y/4 \rfloor$; bars show the trajectory interquartile range.

cumference and the wall average lies closer to two than the full- x average. This is the cleaner statement: the algebraic contribution is carried by the walls, while the full- x estimator retains a short-distance bulk admixture. The persistent left-right offset is visible rather than averaged away. Since only three circumferences retain the full x resolution, we do not extrapolate them.

C. A2: endpoint-packet reduction

Combine only unique, verified endpoint slots. Analyze hard and soft walls separately. For each wall report displacement, fitted velocity, fit quality, packet retention,

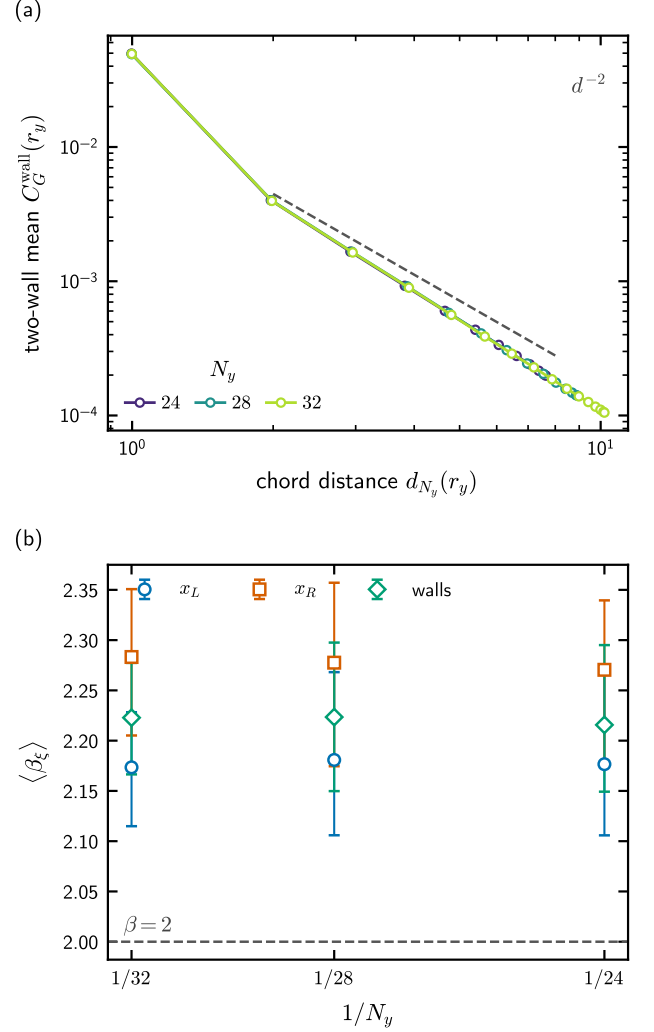


FIG. 2. Wall-resolved version of Fig. 1. (a) Mean of the x_L and x_R correlators, with the critical d^{-2} guide. (b) Exponents at each wall and for their within-trajectory average. Each size contains $S = 100$ independent pure half-filled trajectories evolved to $2N_y$ with hard-wall perfect correction. Fits use $2 \leq r_y \leq \lfloor N_y/4 \rfloor$; bars show the trajectory interquartile range.

source/window sensitivity, and the matched-trivial result.

The circuit data move with wall 0 positive and wall 1 negative, whereas the recent gate imported the opposite absolute orientation from the exact benchmark. This is a sign/protocol reconciliation problem, not an absence of counterpropagation. Tie wall labels to the circuit Chern and coordinate conventions; do not silently flip the sign. Treat poor soft-wall retention as a separate reported limitation.

D. A2b: launched fixed-geometry modular-handedness ensemble

On September 3, 2026, a new trajectory-first CPU campaign was locked and launched from `OO_WORKSPACE/CURRENT/modular_handedness_cpu_pilot/`. It contains 100 independent hard-wall and 100 independent soft-wall trajectories at $N_x = 20$, $N_y = 40$, $n_{\text{shell}} = 1$, with $80 = 2N_y$ physical cycles, perfect correction, pure half filling, the legacy raster- y order, and complex128 arithmetic. The hard construction is support truncated with slab-only dynamics; the soft construction is the untruncated explicit interface. The root seed is 2026090302.

Each deterministic task calls the canonical CPU `classA_U1FGTN.run_markov_circuit`, saves only its final occupied frame, and publishes an NPZ/completion-JSON pair. Modular reduction is a separate resumable stage: all 40 translated half-cylinder cuts are constructed and propagated within each trajectory before any ensemble average. The primary fit is a free-intercept fit on $1 \leq t_{\text{mod}} \leq 5$, with locked fit-window, spectral-cutoff, and wall-window sensitivity analyses. Whole trajectories are bootstrapped with 10,000 fixed-seed draws.

The production session is `modular_handedness.s100` on host `jian1`, pinned to 28 physical cores on NUMA node 0. Its stable output root is `modular_handedness_cpu_pilot/outputs/modular_handedness_nx20_ny40_hard_soft_s100_v1/`. The simulation configuration hash is `f9a49e33e4f8401a4fbae340272cfd5d9b1ca1f6e5ecdd3c1efd61ba04ea14b1`; the analysis configuration hash is `c0042b1bcb3b9aa33b03fa7a8fe6498e2955d2cf470f47c9f68a4a5bb4ec3d836`. The measured completion estimate is 8–10 hours. The absolute wall-Chern orientation remains unresolved, so raw $+y$ velocities stay primary and no post-hoc sign flip is permitted.

E. A3: transverse-width purification audit

Regenerate the conclusion from the completed $N_x = 20, 24, 28$, $N_y = 20$ max-mix data. Report wall and bulk purification times, late wall entropy, uncertainty, and whether a transverse trend is resolved. This is an analysis repair, not a new simulation campaign.

IV. MINIMAL NEW SIMULATION QUEUE

New work begins only after the corresponding analysis is complete, except for the explicitly authorized A2b fixed-geometry campaign:

1. **H1 fixed-geometry ensemble:** the 100 hard plus 100 soft CPU campaign in A2b is launched; resume the same stable output root if interrupted.
2. **Engine equivalence:** replay one small or moderate fixed-seed case with the current canonical GPU

engine and compare the scientific arrays at the appropriate tolerance.

3. **Wall-state top-up:** after A1, select at most one or two geometries targeted at the observed uncertainty or finite-size weakness. Finishing the incomplete $N_y = 60$ archive is a candidate, not an automatic requirement.

There is no current authorization for a fresh P1 campaign, continuation of the retired $L = 36, 40$ uniform-bulk extension, broad purification rerun, or additional large parameter scan beyond A2b.

V. CLAIM-DEPENDENT OPTIONAL WORK

Thermodynamic tangent closing.: Run more sizes and longer time convergence only if the paper claims a thermodynamic zero tangent gap. Otherwise retain the supported finite-size statement.

Higher-Renyi common ensemble.: Measure $q = 2, 3$ entropies and charge fluctuations on a small common ensemble only if precision CFT consistency is central.

Frozen-record stochastic flux.: Run a minimal fixed-record replay only if a branchwise integer index is needed. Exact twist and pump results already provide calibration.

True feedback or phase-noise robustness.: Design a separate campaign only if a quantitative robustness interval becomes a paper claim. Partial postselection is not imperfect feedback.

VI. DEFERRED FROM THE CORE PLAN

The following are not prerequisites for the current central physics:

- full sampled Born modular-response susceptibility, whose score term is currently noise dominated;
- the 5500-trajectory log-Gram α scan;
- many-body operator-CFT or record-CFT towers;
- a universal same-record tangent-purification exponent;
- large phase-noise and feedback-error matrices.

These may become separate projects but should not be presented as validation of an otherwise unsupported wall state.

VII. SIMPLE COLAB EXECUTION CONTRACT

Every new notebook follows the repository policy:

1. one notebook, one runner, the canonical GPU engine, and only required observer modules;
2. code copied from Drive to local `/content` before execution;
3. deterministic configurations or immutable shards of at most five trajectories;
4. an outer `tqdm` task bar and an inner cycle, trajectory, batch, or scan-point bar;
5. local scratch and compact streamed observables;
6. one atomic result plus a completion JSON written last;
7. restart by checksum-verifying completed tasks;
8. for tasks longer than roughly one hour, one rolling checkpoint containing native state, ranks, RNG states, observer accumulators, configuration identity, and completed cycle or point.

The default design excludes a second Drive API path, server leases, generation/pointer protocols, remote-status subprocesses, migration ledgers, session dashboards, and automatic orphan repair. A Drive failure leaves the task pending. Rerunning continues from the last verified task or checkpoint.

VIII. WORKING DECISION LEDGER

IX. OPEN QUESTIONS FOR THE NEXT UPDATE

1. What is the shortest final statement of the paper’s central numerical claim?

TABLE II. Living decisions for the next revision.

Decision	Current position
Bulk P1	Complete; no rerun
Exact B0	Complete; reuse
Wall CFT	Analyze existing joint archive first
H1 chirality	Packet handedness primary; S100 hard plus S100 soft CPU campaign launched; resolve absolute sign
Born response	Remove from core
Soft wall	Report retention limitation before further runs
Tangent	Keep finite-size claim unless more scaling is requested
Stochastic flux	Optional
Operator/record CFT	Deferred
Colab design	Simple task resume plus one rolling checkpoint for long tasks

2. Is consistency with $c = k = 1$ sufficient, or is a precision higher-Renyi/common-ensemble test required?
3. Is a finite-size localized tangent sector sufficient, or must the paper claim a thermodynamic gap closing?
4. Does the paper require a stochastic branchwise integer flux invariant?
5. Should the soft-wall packet construction remain in the main text, move to the supplement as a limitation, or be removed?

Revision rule. Future updates edit this document in place. Completed analyses move into Table I; rejected ideas move to the deferred section with a one-sentence reason; new experiments must name the paper claim they support.